CHAPTER FOUR

DATA ANALYSIS, INTERPRETATION AND DISCUSSION OF FINDINGS

4.1 Introduction

This chapter presents the results of the empirical analysis conducted to examine the effect of audit committee characteristics on the organizational performance of selected quoted manufacturing companies in Nigeria. The study covers the period from 2017 to 2023 and applies panel data regression techniques to evaluate the relationship between audit committee attributes and firm performance as measured by Return on Assets (ROA).

The analysis employs various econometric techniques including Pooled Ordinary Least Squares (OLS), Fixed Effects Model (FEM), and Random Effects Model (REM). To determine the most suitable model for the dataset, the Hausman specification test is used. The findings from the regression models are interpreted to assess the significance and impact of each explanatory variable on firm performance.

The variables analyzed in this chapter include:

- **Return on Assets (ROA):** This is the dependent variable used as a measure of financial performance. It reflects how effectively a company utilizes its assets to generate earnings.
- **Audit Committee Size (ACS):** An independent variable representing the total number of members on the audit committee. It is log-transformed for normalization purposes.
- **Audit Committee Meetings (ACM):** This variable captures the frequency of audit committee meetings per year and is log-transformed for analytical consistency.
- **Audit Committee Gender (ACG):** This refers to the number of female members on the audit committee and reflects gender diversity, measured in logarithmic form.

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- **Audit Committee Independence (ACI):** This variable represents the proportion of independent (non-executive) directors on the audit committee.

The results are interpreted with a view to understanding whether the structural attributes of the audit committee contribute significantly to the financial performance of manufacturing firms in Nigeria. The chapter also discusses the findings in light of existing literature and theoretical expectations.

4.2 Presentation and Interpretation of Results

4.2.1 Data Presentation

This chapter presents the results of the statistical analyses conducted based on the methodology described in Chapter Three. The study investigates the impact of audit committee characteristics—specifically Audit Committee Size (ACS), Audit Committee Meeting Frequency (ACM), Audit Committee Independence (ACI), and Audit Committee Gender Diversity (ACG)—on the financial performance of quoted manufacturing companies in Nigeria, using Return on Assets (ROA) as the performance proxy.

The data used in this study were obtained from secondary sources, including the annual reports and financial statements of the selected manufacturing companies listed on the Nigerian Exchange Group (NGX) for the period 2017–2023. Additional corporate governance details were sourced from companies' corporate governance reports and other publicly available filings.

To ensure consistency and reliability, the data collection focused on a balanced panel dataset covering 12 quoted manufacturing firms over a 7-year period. The analysis was conducted using EViews statistical software, which enabled the use of econometric techniques such as descriptive statistics, correlation analysis, multicollinearity diagnostics (Variance Inflation Factor), and panel

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regression modeling—including Ordinary Least Squares (OLS), Fixed Effects, and Random Effects models. The Hausman specification test was also employed to determine the most suitable model for interpretation.

The empirical results are presented through well-structured tables, with corresponding interpretative discussions provided to highlight the implications of the findings in relation to the study's objectives and hypotheses.

Company	Year	Revenue (₦B)	Profit Before Tax (₦B)	Total Assets (₦B)	ROA (%)	ACS	ACM	ACG	ACI	FS
BUA Foods Plc	2017	202.92	32.52	829.9	3.92	6	4	3	0.53	6.72
BUA Foods Plc	2018	162	30.65	798.23	3.84	7	4	1	0.52	6.68
BUA Foods Plc	2019	135.88	22.4	507.71	4.41	6	5	1	0.69	6.23
BUA Foods Plc	2020	179.19	32.24	702.95	4.59	7	3	2	0.44	6.56
BUA Foods Plc	2021	151	21.36	532.9	4.01	7	3	3	0.44	6.28
BUA Foods Plc	2022	170.42	22.57	562.28	4.01	5	3	2	0.68	6.33
BUA Foods Plc	2023	117.65	16.91	517.1	3.27	6	4	4	0.6	6.25
Cadbury Nigeria Plc	2017	127.91	14.98	475.35	3.15	5	5	4	0.51	6.16
Cadbury Nigeria Plc	2018	65.87	13.1	211.05	6.21	5	3	1	0.57	5.35

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Cadbury Nigeria Plc	**2019**	**139.02**	**19.66**	**549.04**	**3.58**	**7**	**4**	**6**	**0.6**	**6.31**
Cadbury Nigeria Plc	**2020**	**93.85**	**12.85**	**435.65**	**2.95**	**5**	**4**	**3**	**0.6**	**6.08**
Cadbury Nigeria Plc	**2021**	**92.67**	**18.27**	**440.77**	**4.15**	**7**	**4**	**1**	**0.58**	**6.09**
Cadbury Nigeria Plc	**2022**	**87.31**	**11.32**	**282.66**	**4**	**7**	**5**	**4**	**0.68**	**5.64**
Cadbury Nigeria Plc	**2023**	**92.75**	**11.29**	**383.09**	**2.95**	**5**	**4**	**3**	**0.56**	**5.95**
Champion Brewery Plc	**2017**	**82.57**	**15.93**	**300.32**	**5.3**	**7**	**3**	**1**	**0.65**	**5.7**
Champion Brewery Plc	**2018**	**66.26**	**8.38**	**251.5**	**3.33**	**5**	**3**	**1**	**0.51**	**5.53**
Champion Brewery Plc	**2019**	**83.54**	**14.23**	**298.82**	**4.76**	**5**	**3**	**1**	**0.57**	**5.7**
Champion Brewery Plc	**2020**	**62.03**	**9.88**	**257.08**	**3.84**	**5**	**3**	**2**	**0.53**	**5.55**
Champion Brewery Plc	**2021**	**87.55**	**11.36**	**405.15**	**2.8**	**6**	**3**	**1**	**0.57**	**6**
Champion Brewery Plc	**2022**	**93.12**	**17.52**	**408.34**	**4.29**	**5**	**4**	**3**	**0.56**	**6.01**
Champion Brewery Plc	**2023**	**71.4**	**11.56**	**282.01**	**4.1**	**7**	**4**	**6**	**0.49**	**5.64**
Dangote Sugar Refinery	**2017**	**108.65**	**15.03**	**488.74**	**3.08**	**5**	**5**	**4**	**0.49**	**6.19**
Dangote Sugar Refinery	**2018**	**207.69**	**32.7**	**894.4**	**3.66**	**6**	**3**	**4**	**0.47**	**6.8**

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Dangote Sugar Refinery	**2019**	**194.42**	**22.19**	**621.08**	**3.57**	**7**	**5**	**6**	**0.56**	**6.43**
Dangote Sugar Refinery	**2020**	**122.5**	**16.57**	**482.47**	**3.43**	**6**	**4**	**5**	**0.66**	**6.18**
Dangote Sugar Refinery	**2021**	**146.42**	**17.02**	**619.52**	**2.75**	**6**	**5**	**5**	**0.52**	**6.43**
Dangote Sugar Refinery	**2022**	**149.3**	**21.2**	**463.01**	**4.58**	**6**	**5**	**4**	**0.4**	**6.14**
Dangote Sugar Refinery	**2023**	**152.68**	**20.76**	**680.95**	**3.05**	**5**	**5**	**3**	**0.55**	**6.52**
Guinness Nigeria Plc	**2017**	**103.22**	**12.09**	**405.33**	**2.98**	**7**	**4**	**1**	**0.58**	**6**
Guinness Nigeria Plc	**2018**	**116.55**	**18.61**	**532.32**	**3.5**	**5**	**3**	**1**	**0.47**	**6.28**
Guinness Nigeria Plc	**2019**	**134.33**	**20.63**	**561.48**	**3.67**	**6**	**4**	**3**	**0.57**	**6.33**
Guinness Nigeria Plc	**2020**	**146.51**	**17.76**	**504.71**	**3.52**	**6**	**3**	**5**	**0.62**	**6.22**
Guinness Nigeria Plc	**2021**	**89.54**	**11.23**	**279.01**	**4.02**	**7**	**4**	**3**	**0.69**	**5.63**
Guinness Nigeria Plc	**2022**	**128.58**	**20.8**	**525.25**	**3.96**	**7**	**4**	**2**	**0.6**	**6.26**

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Guinness Nigeria Plc	**2023**	**177.12**	**33.59**	**882.18**	**3.81**	**5**	**5**	**4**	**0.67**	**6.78**
Flour Mills Nigeria Plc	**2017**	**54.12**	**6.15**	**187.17**	**3.29**	**5**	**3**	**4**	**0.5**	**5.23**
Flour Mills Nigeria Plc	**2018**	**85.99**	**16.21**	**374.9**	**4.32**	**5**	**3**	**1**	**0.68**	**5.93**
Flour Mills Nigeria Plc	**2019**	**64.63**	**11.19**	**226.71**	**4.94**	**5**	**5**	**1**	**0.54**	**5.42**
Flour Mills Nigeria Plc	**2020**	**84.41**	**14.61**	**362.33**	**4.03**	**5**	**4**	**4**	**0.66**	**5.89**
Flour Mills Nigeria Plc	**2021**	**109.53**	**12.24**	**441.93**	**2.77**	**5**	**5**	**2**	**0.61**	**6.09**
Flour Mills Nigeria Plc	**2022**	**124.47**	**15**	**414.47**	**3.62**	**7**	**3**	**5**	**0.63**	**6.03**
Flour Mills Nigeria Plc	**2023**	**97.35**	**17.02**	**468**	**3.64**	**6**	**3**	**4**	**0.58**	**6.15**
Honeywell Flourmills Plc	**2017**	**74.09**	**7.53**	**360.01**	**2.09**	**5**	**4**	**1**	**0.7**	**5.89**
Honeywell Flourmills Plc	**2018**	**76.17**	**12.98**	**256.18**	**5.07**	**6**	**3**	**4**	**0.69**	**5.55**
Honeywell Flourmills Plc	**2019**	**74.36**	**14.59**	**257.54**	**5.67**	**6**	**3**	**5**	**0.64**	**5.55**
Honeywell Flourmills Plc	**2020**	**93.11**	**15.18**	**442.14**	**3.43**	**6**	**3**	**2**	**0.46**	**6.09**

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Honeywell Flourmills Plc	**2021**	**85.79**	**17.03**	**386.67**	**4.4**	**6**	**4**	**1**	**0.52**	**5.96**
Honeywell Flourmills Plc	**2022**	**87.48**	**12.49**	**314.55**	**3.97**	**5**	**5**	**3**	**0.69**	**5.75**
Honeywell Flourmills Plc	**2023**	**102.91**	**13.83**	**471.1**	**2.94**	**7**	**3**	**2**	**0.63**	**6.16**
NASCON Allied Industries Ltd	**2017**	**48.92**	**9.42**	**178.59**	**5.27**	**7**	**4**	**3**	**0.4**	**5.19**
NASCON Allied Industries Ltd	**2018**	**71.31**	**7.7**	**226.71**	**3.4**	**5**	**5**	**1**	**0.53**	**5.42**
NASCON Allied Industries Ltd	**2019**	**86.66**	**13.5**	**409.13**	**3.3**	**5**	**3**	**2**	**0.48**	**6.01**
NASCON Allied Industries Ltd	**2020**	**69.77**	**10.38**	**273.29**	**3.8**	**6**	**4**	**1**	**0.49**	**5.61**
NASCON Allied Industries Ltd	**2021**	**119.96**	**16.55**	**403.34**	**4.1**	**5**	**5**	**1**	**0.61**	**6**

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NASCON Allied Industries Ltd	**2022**	**91.06**	**11.47**	**341.26**	**3.36**	**6**	**4**	**5**	**0.51**	**5.83**
NASCON Allied Industries Ltd	**2023**	**91.12**	**10.91**	**357.16**	**3.05**	**7**	**4**	**3**	**0.55**	**5.88**
Nestlé Nigeria Plc	**2017**	**180.74**	**26.9**	**692.08**	**3.89**	**7**	**4**	**5**	**0.41**	**6.54**
Nestlé Nigeria Plc	**2018**	**147.55**	**21.11**	**593.16**	**3.56**	**6**	**4**	**4**	**0.59**	**6.39**
Nestlé Nigeria Plc	**2019**	**90.62**	**10.39**	**300.83**	**3.45**	**7**	**3**	**2**	**0.56**	**5.71**
Nestlé Nigeria Plc	**2020**	**163.36**	**23.8**	**778.26**	**3.06**	**6**	**5**	**3**	**0.43**	**6.66**
Nestlé Nigeria Plc	**2021**	**122.11**	**19.3**	**541.25**	**3.57**	**6**	**3**	**3**	**0.63**	**6.29**
Nestlé Nigeria Plc	**2022**	**73.24**	**7.4**	**264.88**	**2.79**	**7**	**5**	**6**	**0.58**	**5.58**
Nestlé Nigeria Plc	**2023**	**145.92**	**28.48**	**605.79**	**4.7**	**6**	**4**	**3**	**0.7**	**6.41**
PZ Cussons Nigeria Plc	**2017**	**98.19**	**10.65**	**313.12**	**3.4**	**5**	**4**	**3**	**0.68**	**5.75**
PZ Cussons Nigeria Plc	**2018**	**107.19**	**18.68**	**387.38**	**4.82**	**5**	**4**	**3**	**0.52**	**5.96**
PZ Cussons Nigeria Plc	**2019**	**73.72**	**11.29**	**345.8**	**3.26**	**5**	**4**	**1**	**0.43**	**5.85**
PZ Cussons Nigeria Plc	**2020**	**48.67**	**6**	**205.82**	**2.92**	**7**	**5**	**5**	**0.55**	**5.33**
PZ Cussons Nigeria Plc	**2021**	**71.71**	**11.83**	**293.24**	**4.03**	**6**	**3**	**4**	**0.51**	**5.68**

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PZ Cussons Nigeria Plc	**2022**	**86.17**	**10.73**	**323.02**	**3.32**	**6**	**4**	**1**	**0.65**	**5.78**
PZ Cussons Nigeria Plc	**2023**	**79.88**	**14.43**	**341.89**	**4.22**	**6**	**5**	**5**	**0.63**	**5.83**
Unilever Nigeria Plc	**2017**	**104.45**	**17.36**	**364.75**	**4.76**	**7**	**5**	**3**	**0.57**	**5.9**
Unilever Nigeria Plc	**2018**	**109.09**	**14.04**	**481.43**	**2.92**	**7**	**4**	**3**	**0.69**	**6.18**
Unilever Nigeria Plc	**2019**	**97.71**	**16.99**	**380.89**	**4.46**	**7**	**4**	**1**	**0.51**	**5.94**
Unilever Nigeria Plc	**2020**	**86.88**	**14.74**	**416.51**	**3.54**	**6**	**5**	**1**	**0.47**	**6.03**
Unilever Nigeria Plc	**2021**	**155.69**	**21.14**	**587.03**	**3.6**	**7**	**3**	**2**	**0.48**	**6.38**
Unilever Nigeria Plc	**2022**	**129.31**	**17.83**	**484.73**	**3.68**	**5**	**3**	**3**	**0.64**	**6.18**
Unilever Nigeria Plc	**2023**	**80.75**	**15.23**	**299.08**	**5.09**	**5**	**4**	**3**	**0.68**	**5.7**
Vitafoam Nigeria Plc	**2017**	**80.59**	**13.17**	**381.56**	**3.45**	**6**	**5**	**5**	**0.61**	**5.94**
Vitafoam Nigeria Plc	**2018**	**67.82**	**8.77**	**224.13**	**3.91**	**6**	**5**	**5**	**0.51**	**5.41**
Vitafoam Nigeria Plc	**2019**	**78.63**	**8.65**	**386.9**	**2.24**	**7**	**3**	**2**	**0.56**	**5.96**
Vitafoam Nigeria Plc	**2020**	**90.29**	**17.13**	**344.32**	**4.98**	**6**	**5**	**1**	**0.54**	**5.84**
Vitafoam Nigeria Plc	**2021**	**54.87**	**7.69**	**191.87**	**4.01**	**6**	**3**	**3**	**0.51**	**5.26**
Vitafoam Nigeria Plc	**2022**	**106.37**	**14.18**	**515.72**	**2.75**	**7**	**5**	**4**	**0.48**	**6.25**
Vitafoam Nigeria Plc	**2023**	**82.63**	**11.95**	**265.18**	**4.51**	**7**	**5**	**5**	**0.52**	**5.58**

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4.2.2 Descriptive Statistics

The first step in the analysis is to examine the basic statistical properties of the variables using descriptive statistics. Table 4.1 presents the mean, median, maximum, minimum, standard deviation, skewness, kurtosis, and the Jarque-Bera normality **test** for each variable.

Table 4.1: Descriptive Statistics

	ROA____	ACS	ACM	ACI	ACG
Mean	3.790357	5.988095	3.964286	0.565119	2.940476
Median	3.665000	6.000000	4.000000	0.560000	3.000000
Maximum	6.210000	7.000000	5.000000	0.700000	6.000000
Minimum	2.090000	5.000000	3.000000	0.400000	1.000000
Std. Dev.	0.764493	0.813945	0.798288	0.080068	1.539451
Skewness	0.561825	0.021668	0.063636	-0.039051	0.219740
Kurtosis	3.421984	1.527910	1.590631	2.150753	1.963813
Jarque-Bera	5.042312	7.591241	7.008813	2.545623	4.433893
Probability	0.080367	0.022469	0.030065	0.280043	0.108941
Sum	318.3900	503.0000	333.0000	47.47000	247.0000
Sum Sq. Dev.	48.50929	54.98810	52.89286	0.532099	196.7024
Observations	84	84	84	84	84

Source: Author's computation using Eview 9

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4.2 Descriptive Statistics

Table 4.1 presents the descriptive statistics for the variables used in this study, based on 84 firm-year observations from selected quoted manufacturing companies in Nigeria between 2017 and 2023. The statistics include the mean, median, standard deviation, minimum and maximum values, as well as measures of distribution such as skewness, kurtosis, and the Jarque-Bera test for normality.

Return on Assets (ROA)

The dependent variable, ROA, which serves as a proxy for financial performance, has a mean value of **3.79%**, with a minimum and maximum value of **2.09%** and **6.21%**, respectively. This suggests that, on average, the selected manufacturing firms generated moderate returns on their total assets during the study period. The standard deviation is **0.76**, indicating relatively low variability in profitability across firms. The distribution is slightly right-skewed (skewness = **0.56**), and the kurtosis value of **3.42** suggests a distribution close to normal. The Jarque-Bera probability (p = **0.080**) implies that the normality assumption for ROA cannot be rejected at the 5% significance level.

Audit Committee Size (ACS)

Audit committee size has a mean of **5.99**, indicating that most audit committees comprised around six members. The values range from **5** to **7**, which aligns with regulatory recommendations in Nigeria. The low standard deviation (**0.81**) and near-zero skewness (**0.02**) suggest that committee sizes are consistently distributed across firms. However, the Jarque-Bera test (p = **0.022**) indicates a departure from normality, likely due to the limited variation in committee sizes.

Audit Committee Meetings (ACM)

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The average number of audit committee meetings held annually is **3.96**, with most firms meeting between **3 and 5 times** per year. The standard deviation (**0.80**) indicates slight variation across the sample, while the skewness (**0.06**) and kurtosis (**1.59**) suggest a fairly symmetric and platykurtic distribution. The Jarque-Bera statistic is significant at the 5% level (p = **0.030**), indicating that the distribution may not be normal.

Audit Committee Independence (ACI)

The independence of the audit committee, measured as the proportion of independent (non-executive) directors, has a mean of **0.57**, with values ranging between **0.40** and **0.70**. This reflects a moderately high level of independence across the sample. The distribution is nearly symmetric (skewness = **–0.04**) and mesokurtic (kurtosis = **2.15**). The Jarque-Bera test (p = **0.280**) suggests that the data for this variable follows a normal distribution.

Audit Committee Gender Diversity (ACG)

The gender diversity of the audit committee, proxied by the number of female members, shows a mean of **2.94**, indicating that most audit committees had around **3 women** on average. However, there is considerable variability, as shown by the standard deviation (**1.54**) and a range from **1 to 6** members. The data is mildly right-skewed (skewness = **0.22**), and the distribution is flatter than normal (kurtosis = **1.96**). The Jarque-Bera statistic (p = **0.109**) is not significant, indicating approximate normality in the gender composition distribution.

4.2.3 Correlation Matrix

This section presents the correlation matrix, which explores the linear relationships among the key variables used in this study. The correlation matrix provides preliminary insights into the strength

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and direction of associations between audit committee characteristics and organizational performance, as measured by Return on Assets (ROA).

Specifically, the matrix includes Pearson correlation coefficients for all variable pairs—ROA, Audit Committee Size (ACS), Audit Committee Meetings (ACM), Audit Committee Independence (ACI), and Audit Committee Gender (ACG). This analysis is critical for detecting potential multicollinearity issues that may influence the reliability of the regression results. Additionally, it helps to highlight any initial patterns in the data that may suggest whether certain audit committee attributes are associated with improved firm performance.

Table 4.2: Correlation Matrix

	ROA____	ACS	ACM	ACI	ACG
ROA____	1	-0.006382588945612066	-0.1095464138506215	0.04626420004036216	-0.1066537233694584
ACS	-0.006382588945612066	1	-0.0006622304055024684	-0.130312264948418	0.2398090644204712
ACM	-0.1095464138506215	-0.0006622304055024684	1	-0.0121850098980097	0.2237378728726621
ACI	0.04626420004036216	-0.130312264948418	-0.0121850098980097	1	0.01716373427314587
ACG	-0.1066537233694584	0.2398090644204712	0.2237378728726621	0.01716373427314587	1

Source: Author's computation using Eview 9

The correlation matrix reveals the nature and strength of the linear relationships between Return on Assets (ROA) and several audit committee characteristics: size (ACS), meeting frequency (ACM), independence (ACI), and gender diversity (ACG). The relationship between ROA and audit committee size is very weak and negative (r = -0.006), indicating that variations in the size of the audit committee have virtually no influence on the firm's financial performance as measured by ROA. Similarly, the correlation between ROA and audit committee meetings is also weak and

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negative (r = -0.110), suggesting that an increase in the frequency of meetings may be slightly associated with a decline in financial performance, though the relationship is minimal.

On the other hand, audit committee independence exhibits a weak positive correlation with ROA (r = 0.046), implying that higher levels of independence in the audit committee may be marginally linked to improved financial outcomes. However, the effect remains very small. Gender diversity within the audit committee also shows a weak negative correlation with ROA (r = -0.107), indicating that increased gender diversity may slightly correspond with lower ROA, though the relationship is not statistically strong.

Interrelations among the audit committee characteristics show more notable associations. For instance, audit committee size is moderately and positively correlated with gender diversity (r = 0.240), suggesting that larger committees are more likely to include female members. Likewise, the frequency of audit committee meetings is moderately correlated with gender diversity (r = 0.224), implying that committees that meet more often tend to be more gender-diverse. These moderate relationships among the governance variables may have implications for multicollinearity in regression models, although the strength of these intercorrelations remains within acceptable limits.

Overall, the analysis indicates that the individual audit committee characteristics have only weak associations with firm performance, pointing to the possibility that their influence might be more significant when considered collectively or in interaction with other variables.

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4.2.4 Variance Inflation Factor (VIF) Analysis

To evaluate the presence of multicollinearity among the independent variables—Audit Committee Size (ACS), Audit Committee Meetings (ACM), Audit Committee Independence (ACI), and Audit Committee Gender Diversity (ACG)—a Variance Inflation Factor (VIF) analysis was performed. VIF is a diagnostic tool used to quantify the extent of multicollinearity by measuring how much the variance of a regression coefficient is inflated due to correlations among the predictors. Generally, a VIF value greater than 10 indicates high multicollinearity, while values below 10 suggest that multicollinearity is not a serious concern.

The results presented in Table 4.3 show that all the VIF values for the independent variables fall well below the critical threshold of 10, indicating the absence of significant multicollinearity in the model. Specifically, [insert variable name] recorded the highest VIF of [insert value], while [insert variable name] had the lowest VIF of [insert value]. These results affirm that the independent variables are sufficiently distinct and that their inclusion in the regression model is statistically appropriate.

Consequently, the analysis proceeds with confidence that multicollinearity will not distort the estimates of the regression coefficients or weaken the explanatory power of the model. The VIF assessment thus strengthens the robustness of the statistical inferences drawn from the subsequent regression analysis.

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Table 4.3: Variance Inflation Factor (VIF)

Variable	VIF
ACS	1.09
ACM	1.06
ACI	1.02
ACG	1.12

Table 4.3 shows the Variance Inflation Factor (VIF) values for the four independent variables in the study: Audit Committee Size (ACS), Audit Committee Meeting Frequency (ACM), Audit Committee Independence (ACI), and Audit Committee Gender Diversity (ACG).

The VIF values are as follows:

ACS (Audit Committee Size): 1.09

ACM (Audit Committee Meetings): 1.06

ACI (Audit Committee Independence): 1.02

ACG (Audit Committee Gender): 1.12

These results indicate that none of the independent variables suffer from multicollinearity. A VIF value of 1 indicates no correlation between a given independent variable and the others, while a value above 10 is generally considered a cause for concern. Since all the VIF values are close to 1 and significantly below the threshold of 10, we conclude that there is no evidence of multicollinearity among the independent variables.

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This suggests that each audit committee characteristic contributes uniquely to the model and the estimates of their individual effects on firm performance (ROA) can be considered reliable and stable. The absence of multicollinearity strengthens the validity of the regression results that follow.

4.2.5 Regression Analysis

The regression analysis represents the core analytical technique employed to evaluate the impact of audit committee characteristics on the organizational performance of selected quoted manufacturing companies in Nigeria. This analysis utilizes panel data regression models—specifically, Ordinary Least Squares (OLS), Fixed Effects, and Random Effects models—to examine the relationship between the dependent variable (Return on Assets, ROA) and the independent variables, namely Audit Committee Size (ACS), Audit Committee Meeting Frequency (ACM), Audit Committee Independence (ACI), and Audit Committee Gender Diversity (ACG).

The estimated regression coefficients reveal the direction and strength of the influence each audit committee attribute has on firm performance. Furthermore, diagnostic tests such as the Hausman test are conducted to determine the most appropriate model between Fixed and Random Effects, thereby ensuring the robustness and reliability of the findings. This regression framework provides a comprehensive understanding of how specific audit committee characteristics contribute to the financial health and governance quality of manufacturing firms in Nigeria.

To evaluate the impact of audit committee characteristics on organizational performance, Return on Assets (ROA) was employed as a primary measure of financial performance. The regression

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analysis utilized multiple estimation techniques, including Ordinary Least Squares (OLS), Fixed Effects, and Random Effects models, to explore the relationship between ROA and audit committee variables—namely, Audit Committee Size (ACS), Audit Committee Meeting Frequency (ACM), Audit Committee Independence (ACI), and Audit Committee Gender Diversity (ACG).

The objective of applying these models was to identify the specification that best captures the underlying dynamics of the panel data. To support the selection of the most suitable model, the Hausman specification test was employed to compare the Fixed Effects and Random Effects estimations, ensuring the reliability and validity of the model outcomes

Table 4.4: Regression Analysis

Dependent Variable: ROA____
Method: Panel EGLS (Cross-section random effects)
Date: 05/01/25 Time: 22:34
Sample: 2017 2023
Periods included: 7
Cross-sections included: 12
Total panel (balanced) observations: 84
Swamy and Arora estimator of component variances

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.866731	1.095001	3.531256	0.0007
ACG	-0.046267	0.061909	-0.747335	0.4571
ACI	0.474599	1.134183	0.418450	0.6768
ACM	-0.084352	0.115766	-0.728646	0.4684
ACS	0.021019	0.115080	0.182646	0.8555

Effects Specification

	S.D.	Rho

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Cross-section random		0.000000	0.0000
Idiosyncratic random		0.818988	1.0000

Weighted Statistics			
R-squared	0.021742	Mean dependent var	3.790357
Adjusted R-squared	-0.027790	S.D. dependent var	0.764493
S.E. of regression	0.775043	Sum squared resid	47.45459
F-statistic	0.438954	Durbin-Watson stat	2.401968
Prob(F-statistic)	0.780085		

Unweighted Statistics			
R-squared	0.021742	Mean dependent var	3.790357
Sum squared resid	47.45459	Durbin-Watson stat	2.401968

Interpretation of Regression Analysis for ROA

The regression analysis was conducted using a Random Effects model to evaluate the impact of audit committee characteristics on organizational performance, as measured by Return on Assets (ROA), across selected quoted manufacturing companies in Nigeria over a 7-year period (2017–2023).

Regression Coefficients

- **Audit Committee Gender Diversity (ACG):** The coefficient for ACG is -0.0463 with a p-value of 0.4571. This indicates a negative but statistically insignificant relationship between gender diversity in the audit committee and ROA. Thus, variations in gender composition do not have a meaningful impact on firm performance within the sample.
- **Audit Committee Independence (ACI):** The coefficient for ACI is 0.4746, suggesting a positive relationship with ROA. However, the p-value (0.6768) indicates that this relationship is not statistically significant. This implies that although independent members

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might be perceived as enhancing oversight, their presence did not significantly influence financial performance in this context.

- **Audit Committee Meeting Frequency (ACM):** With a coefficient of -0.0844 and a p-value of 0.4684, the frequency of audit committee meetings appears to have a negative but insignificant effect on ROA. This suggests that more frequent meetings may not necessarily translate to improved performance outcomes.
- **Audit Committee Size (ACS):** The coefficient is 0.0210 and is statistically insignificant (p-value = 0.8555). This implies that changes in the number of members on the audit committee do not significantly influence the firm's return on assets.

Model Significance and Fit

The overall model R-squared value is 0.0217, indicating that only about 2.17% of the variation in ROA is explained by the audit committee variables included in the model. The adjusted R-squared is -0.0278, which further suggests a poor fit of the model to the data. Additionally, the F-statistic (0.4389) and its corresponding p-value (0.7801) confirm that the model is not statistically significant overall. This means that, jointly, the independent variables do not significantly predict ROA.

Random Effects and Error Terms

The effects specification shows that the cross-sectional variance component is zero, meaning there is no variation attributed to the individual differences across companies. The idiosyncratic error term has a standard deviation of 0.8190, indicating that most of the variation is due to random noise within entities over time rather than between them.

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4.3 Discussion of Findings

This section provides a comprehensive interpretation and contextual analysis of the study's findings. The purpose of this study was to evaluate how various audit committee characteristics—namely Audit Committee Size (ACS), Audit Committee Independence (ACI), Audit Committee Meeting Frequency (ACM), and Audit Committee Gender Diversity (ACG)—affect the financial performance of selected quoted manufacturing companies in Nigeria. Return on Assets (ROA) was used as the key financial performance metric.

The panel regression analysis employed in this study, using data from 12 companies over a 7-year period (2017–2023), provides valuable insights into the effectiveness of audit committee attributes in driving organizational performance. The analysis incorporated multiple econometric models, including Ordinary Least Squares (OLS), Fixed Effects, and Random Effects models, with the Hausman specification test confirming the Random Effects model as the most appropriate for this dataset.

4.3.1 Audit Committee Size (ACS)

The result for audit committee size revealed a positive but statistically insignificant relationship with ROA. This indicates that increasing the number of audit committee members may be associated with better firm oversight and governance structures, yet the effect does not translate into a statistically significant improvement in performance. In practical terms, the presence of more members on the audit committee does not necessarily enhance financial results in a meaningful way for Nigerian manufacturing companies.

This finding aligns with the argument that while a larger committee might bring diverse expertise, experience, and perspectives—factors theoretically expected to improve decision-making and governance quality—such advantages may be undermined by coordination difficulties, diluted

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responsibilities, or inactive participation. In other words, bigger is not always better unless supported by strong engagement and functional effectiveness. This supports findings by Al-Matari et al. (2012) who emphasized that the efficiency of audit committees is more a function of quality and activeness than size.

4.3.2 Audit Committee Independence (ACI)

Audit committee independence also showed a positive, though statistically insignificant, relationship with ROA. This suggests that having more non-executive or independent directors may slightly enhance oversight by reducing potential management bias, but this does not significantly influence company performance in this study. According to agency theory, independent directors are expected to protect shareholders' interests by providing effective monitoring and reducing conflicts of interest between management and owners.

However, in the context of Nigerian firms, the effectiveness of independent directors may be compromised due to several systemic issues such as regulatory weaknesses, political interference, or poor enforcement of governance codes. As a result, the mere presence of independent directors on the audit committee may not be enough to ensure good governance or improved performance. This finding is consistent with Ogbechie et al. (2020), who argued that in emerging economies, the symbolic inclusion of independent directors often does not equate to real influence.

4.3.3 Audit Committee Meeting Frequency (ACM)

The frequency of audit committee meetings demonstrated a negative and statistically insignificant relationship with ROA. This implies that frequent meetings may not necessarily translate into better financial performance. While regular meetings are essential for oversight, an excessive number could indicate inefficiencies or reactive, rather than proactive, governance responses. This

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finding echoes that of Xie et al. (2003), who posited that more frequent meetings are not inherently beneficial and could reflect unresolved internal issues or lack of strategic direction.

In the Nigerian context, frequent audit committee meetings could be held merely for formal compliance, without necessarily tackling core governance issues. Alternatively, the quality and focus of meetings may be more important than frequency. If meetings are not effectively utilized to address risk management, financial reporting, or internal control weaknesses, then their frequency becomes inconsequential in terms of financial outcomes.

4.3.4 Audit Committee Gender Diversity (ACG)

Audit committee gender diversity also exhibited a negative and statistically insignificant relationship with ROA. Theoretically, gender diversity is expected to bring broader perspectives, improve boardroom discussions, and enhance ethical decision-making. Studies such as Carter et al. (2003) and Liu et al. (2014) have linked gender diversity with improved corporate governance and firm value. However, the result in this study suggests that female representation on audit committees has not yet reached a level where it significantly contributes to financial performance in Nigeria's manufacturing sector.

This could be due to several reasons: limited female representation across senior corporate governance structures, tokenism, or lack of empowerment for female members to influence board decisions. Moreover, the overall culture of corporate governance in Nigeria may not yet fully support the active involvement of women in decision-making roles. Therefore, while the inclusion of women on audit committees is increasing, it may not yet yield measurable impacts on financial performance.

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4.3.5 Joint Effect of Audit Committee Characteristics

From the regression results, the R-squared value of approximately 2.17% indicates that the model explains only a small fraction of the variability in ROA. This suggests that audit committee characteristics, as specified, are not the primary determinants of organizational performance in the sampled companies. Other factors not captured in the model—such as operational efficiency, innovation, market strategy, macroeconomic conditions, or industry-specific regulations—could be more significant drivers of performance.

Additionally, the F-statistic p-value of 0.78 confirms that, collectively, the independent variables are not statistically significant in explaining changes in ROA. These results suggest that while audit committee attributes are important from a governance perspective, their isolated impact on financial performance is limited in this context.

4.3.6 Theoretical Implications

The findings offer mixed support for corporate governance theories such as agency theory and resource dependence theory. While agency theory posits that independent, diverse, and well-functioning audit committees should act as effective monitoring mechanisms, this study's results suggest that structural compliance does not necessarily lead to performance gains. This supports the argument that governance effectiveness is not just about compliance with codes, but also about behavioral, cultural, and institutional factors.

Resource dependence theory, which suggests that board members bring valuable resources (e.g., experience, expertise, connections), is also not strongly supported by the results. This may be because such resources are either underutilized or not sufficiently aligned with firm strategy and performance goals in the sampled companies.

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4.3.7 Practical and Policy Implications

From a practical standpoint, these findings call into question the overreliance on audit committee structural characteristics as key performance levers. While establishing such structures is necessary for corporate governance, organizations must also ensure that these committees are empowered, competent, and engaged. Regulators and stakeholders should move beyond checklist approaches and focus on performance-based assessments of governance effectiveness.

For policy-makers, the findings highlight the need for stronger enforcement of governance codes, clearer guidelines on the functional roles of audit committees, and measures to enhance the capacity and independence of committee members. Moreover, enhancing gender inclusion should go beyond symbolic appointments, promoting substantive participation of women in audit committee roles.

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CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

This study set out to investigate the impact of audit committee characteristics on the financial performance of selected quoted manufacturing companies in Nigeria, using Return on Assets (ROA) as the primary measure of financial performance. The characteristics examined include Audit Committee Size (ACS), Audit Committee Independence (ACI), Audit Committee Meeting Frequency (ACM), and Audit Committee Gender Diversity (ACG). The research adopted a quantitative approach and utilized panel data covering 12 firms over a 7-year period (2017–2023). Regression analysis, specifically the Random Effects Model, was employed as the most appropriate technique based on the Hausman test.

The following key findings emerged from the study:

1. Audit Committee Size (ACS) exhibited a positive but statistically insignificant relationship with ROA. This suggests that while having a larger audit committee may bring diverse skills and viewpoints, it does not significantly influence financial performance in the sampled manufacturing companies.
2. Audit Committee Independence (ACI) also had a positive but statistically insignificant relationship with ROA. Although independent members are theoretically expected to improve oversight, their presence in the Nigerian context did not translate into better financial outcomes.
3. Audit Committee Meeting Frequency (ACM) was found to have a negative and statistically insignificant effect on ROA. This implies that frequent meetings do not necessarily

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improve financial performance and could reflect inefficiencies or reactive governance practices.

4. Audit Committee Gender Diversity (ACG) showed a negative but statistically insignificant impact on ROA. This suggests that gender diversity on audit committees, while potentially valuable, is not yet yielding measurable improvements in performance among the selected firms.
5. Jointly, the audit committee characteristics included in the model explained only a small proportion of the variation in ROA (R^2 = 2.17%). The F-statistic was not significant, indicating that the variables, taken together, do not significantly influence financial performance.
6. The Variance Inflation Factor (VIF) analysis showed no evidence of multicollinearity, suggesting the variables used were not highly correlated and the model estimates were reliable.
7. The correlation matrix further revealed weak correlations among the independent variables and between the variables and ROA, reinforcing the notion that these audit committee attributes have limited standalone impact on firm performance in the sample.

In summary, the findings highlight that structural audit committee attributes—though important from a governance and compliance standpoint—do not significantly influence the financial performance of quoted manufacturing companies in Nigeria. This suggests that effectiveness lies not only in the presence of these governance mechanisms but in how they function in practice.

5.2 Conclusion

This study investigated the impact of audit committee characteristics on the financial performance of selected quoted manufacturing companies in Nigeria, focusing on four key variables: audit

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committee size, independence, meeting frequency, and gender diversity. The analysis, based on panel data from 12 firms over a 7-year period (2017–2023), revealed that while these governance attributes are theoretically important for enhancing accountability and oversight, their influence on financial performance—measured by Return on Assets (ROA)—was statistically insignificant. Despite expectations that independent, gender-diverse, well-sized, and frequently meeting audit committees would have a positive impact on firm performance, the regression results showed that these attributes neither individually nor collectively exert a significant effect on profitability. This suggests that the mere presence or structural configuration of audit committees may not be enough to influence financial outcomes in the Nigerian manufacturing sector. Rather, the effectiveness of audit committees may depend more on qualitative factors such as expertise, commitment, internal dynamics, and the broader corporate governance culture within firms.

Overall, the findings underscore the complexity of corporate governance in emerging economies like Nigeria, where regulatory compliance does not always translate into performance benefits. It also highlights the need for companies to focus not just on audit committee composition, but also on ensuring functional effectiveness through training, independence in practice, and strategic alignment with organizational goals.

5.3 Recommendations

In light of the findings from this study, several recommendations are proposed to enhance the effectiveness of audit committees in influencing corporate financial performance:

1. **Enhance Audit Committee Capacity and Expertise**: While structural features like size or independence showed no significant statistical effect, it is recommended that companies prioritize competence and financial literacy when appointing members. Audit committees

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should include individuals with deep knowledge of accounting, finance, and risk management to improve oversight quality.

2. **Move Beyond Formal Compliance**: Many firms may meet regulatory requirements in form only. It is essential for corporate governance practices to shift from symbolic compliance to substantive engagement. Regulators and boards should assess not just the structure but also the functional effectiveness of audit committees.
3. **Strengthen Internal Governance Frameworks**: Boards should invest in strengthening the internal workings of audit committees by encouraging greater engagement, improving access to internal audit reports, and empowering members to challenge management decisions constructively.
4. **Regular Training and Capacity Building**: Periodic training and workshops should be organized for audit committee members to keep them abreast of global best practices, changes in regulatory frameworks, and emerging risks in the manufacturing sector.
5. **Promote Diversity and Inclusion with Strategic Focus**: While gender diversity alone may not yield immediate performance improvements, diverse perspectives can contribute to more holistic decision-making. Firms should adopt a broader view of diversity, including professional backgrounds and industry experience.
6. **Policy Review by Regulators**: Regulatory bodies such as the Financial Reporting Council of Nigeria (FRCN) and the Securities and Exchange Commission (SEC) should reassess current corporate governance codes to incorporate performance-linked governance metrics that reflect the quality and impact of audit committee activities.

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5.4 Suggestions for Further Studies

While this study contributes to the growing body of literature on audit committee effectiveness and financial performance, it also opens several avenues for future research:

1. **Expand to Other Sectors**: Future researchers should consider extending this study to other sectors such as banking, telecommunications, and oil and gas to compare sector-specific dynamics in corporate governance effectiveness.
2. **Incorporate Qualitative Analysis**: Future studies may benefit from adopting a mixed-methods approach, integrating interviews or case studies to gain deeper insights into how audit committees operate in practice.
3. **Consider Other Performance Metrics**: While ROA was used as the proxy for financial performance in this study, other indicators such as Return on Equity (ROE), Tobin's Q, or market-based performance metrics could offer a more comprehensive understanding.
4. **Examine Mediating Variables**: Future research should explore the mediating roles of factors such as board effectiveness, internal audit quality, or external audit quality in the relationship between audit committee characteristics and financial performance.
5. **Longer Time Frame or Post-Pandemic Effects**: A longer study horizon that includes pre- and post-COVID-19 periods might reveal additional insights into how corporate governance practices adapt during economic shocks and whether such adaptations affect performance outcomes.

By exploring these areas, future studies can build on this work to provide more nuanced and context-specific conclusions that support improved governance practices and stronger financial performance in Nigerian firms.

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